

MERGE 4 DESIGN LLC × L.E.E.P., INC.

FORWARD HOME BUILDING SYSTEM

Project Specification — Four-Tier Performance Framework + Factory Assembly Program

Code-Compliant · Net-Zero · Net-Positive · Off-Grid Harvest · Factory-Assembled | City of Wilmington Forward Home Program
Submitted: June 2026 · Companion files: M4D_ForwardHome_Proforma_May2026.xlsx · M4Di_ForwardSystem_AssemblyCost.html

2A. Wall Assembly — Revised: This Revision

This revision integrates the wall assembly cost intelligence tool (M4Di Forward System Assembly Cost Comparator, June 2026) into the full four-tier specification. Section 2B (below) is a new wall assembly matrix showing each layer from interior finish to exterior cladding, the SAPK connector kit by tier, and the installed cost comparison against equivalent stick-frame construction. Sections 9 and 10 are new additions: factory assembly enhancements and the recommended efficiency program for windows, doors, hardware, fixtures, and finishes.

2B. Wall Assembly Matrix — Interior to Exterior by Tier

The following table defines the complete wall assembly for each Forward Home tier, from interior finish to exterior cladding. All SAPK connector products carry existing US ICC-ES listings. No new UL or ICC testing is required for the specified assembly. Costs are ROM ±20%, Mid-Atlantic (Wilmington DE), June 2026, per SF of gross wall area including materials and labor in place.

Element	T1 — Code+	T2 — Zero	T3 — Positive	T4 — Harvest
ZONE 1 — FOUNDATION INTERFACE (inside → outside)				
Sill anchor / base track	Simpson MASOZ mudsill anchor · ESR-2555 · CFS base track AISI S240	Same + SIGA TEAKstrip sill capillary break	Same as T2	Same as T2
Foundation wall	CMU / poured 3,500 psi · R-10 EPS under slab · R-10 edge · 15-mil VR	Superior Walls Xi Plus · TER 2211-02 · 4½" EPS+½" polyiso built-in · no dampproofing	Same as T2	Same as T2
ZONE 2 — WALL PANEL ASSEMBLY (inside → outside)				
Interior finish	½" Type X GWB — factory hung at hub · tape+mud on site	Same · fluid-applied at all electrical penetrations	Same + Pro Clima Intello+ airtight membrane at junctions	Same as T3
CFS spine (where applicable)	600S162-54 · AISI S240 · ESR-4205 · point load transfer at openings + hold-downs	Same	Same	Same
Structural wall panel	ACME 6.5" SIP — Neopor GPS / OSB faces · R-23 · ESR-4689 OR 2×6 SPF-S @ 16" OC + R-23 blown fiberglass	Same SIP · R-23 Rockwool Comfortbatt mineral wool cavity upgrade	Same SIP	Same SIP
WRB + CI layer	Barricade Thermo-Brace SIB R-5 · TER 1507-09 WRB integrated both faces · GREENGUARD Gold	Same + fluid-applied Prosoco R-Guard at all penetrations	Same + continuous air barrier lapped at foundation + roof	Same — critical for island-mode battery efficiency

Element	T1 — Code+	T2 — Zero	T3 — Positive	T4 — Harvest
Outboard CI (T3–T4 only)	—	—	DuPont ArmorWall Plus FR SIS 2¼" R-15 MgO/PUR · ESL-1302 · NFPA 285 · direct cladding attach	ArmorWall Plus FR SIS 3¼" R-21 MgO/PUR · ESL-1302 · max CI tier
Party wall / fire separation	8" CMU continuous foundation → underside SIP roof · inherent STC ≥ 50	Same + Thermafiber 4" + Hilti FS-ONE MAX · STC ≥ 55	Same + Huber EXACOR MGO face · ESR-4635 · 2-hr E119 · STC/IIC ≥ 60	Same as T3
CLT-to-CFS bearing connector	Rothoblaas ALUMEGA concealed · ESR-4546 / AC233 Rothoblaas HBS PLATE screw · ETA- 11/0030	Same	Same — C4 corrosion: ALUMEGA + HBS PLATE EVO / AC257	Same — A4 stainless: ALUMEGA + HBS PLATE A4
Hold-downs	Simpson HTT4 · ESR- 1188	Simpson HTT5 · ESR- 1188	Simpson HTT5 · ESR- 1188	Simpson HTT22 · ESR- 1188
SIP joint spline + airtight tape	ACME manufacturer spline · per ESR-4689 Rothoblaas SIGA Rissan airtight tape at all SIP joints	Same + SIGA FENTRIM IS at window/door perimeters	Same + SIGA MAJREX smart vapor retarder inboard	Same as T3
Stair cassette connector	Simpson SCLC stair cassette ledger · C-C- 2024 · hub-built cassette lowered on site	Same	Same	Same
ZONE 3 — EXTERIOR CLADDING (outside face)				
Primary cladding	Allura® Lap 7" exposure · factory primed · 50-yr warranty 1" ventilated rainscreen on SIB face	Allura® Panel smooth or cedar · Allura Spectrum pre-finished Stainless rainscreen clips · no field paint	Allura® Panel + Shakes accent zones · Spectrum pre-finished Direct attach to ArmorWall MgO face — no hat channel	Same as T3
Trim + soffit	Allura Trim at all openings	Allura Soffit at all overhangs + Allura Trim	Same as T2	Same as T2
ZONE 4 — SAPK CONNECTOR KIT SUMMARY				
Foundation zone	SST MASOZ G90 + HTT4	SST MASOZ + HTT5 + SIGA TEAKstrip	Same as T2	SST MASOZ + HTT22
Wall zone	Rtb ALUMEGA + HBS PLATE #14 SD screw AISI S240	Same + Rtb SIGA Rissan tape	Same + Rtb HBS PLATE EVO + SIGA FENTRIM IS	Same — all A4 stainless
Stair cassette	SST SCLC + SDS screws	Same	Same	Same
All items carry existing US listings	No new UL or ICC testing required	AISI S240, ESR-1188, ESR-2555, ESR-4205, ESR-4546, AC233, IAPMO ER-192	Same + AC257 (EVO corrosion)	Same — A4 SS coating spec
ZONE 5 — EFFECTIVE WALL PERFORMANCE				
Effective R-value	R-21 to R-23	R-22 to R-24	R-38+ (SIP + ArmorWall 2¾")	R-44+ (SIP + ArmorWall 3¾")
Airtightness target	<2.0 ACH50 (auto — SIP)	<1.0 ACH50 — verified mid + CO	<0.6 ACH50 — PHIUS verified	<0.6 ACH50 — island- mode critical
INSTALLED COST COMPARISON — \$/SF WALL (Mid-Atlantic, June 2026 ROM)				
Traditional stick-frame equivalent	\$38.50/SF	\$52.80/SF	\$71.40/SF	\$84.60/SF
Forward System — partial factory	\$36.20/SF	\$44.60/SF	\$55.80/SF	\$62.40/SF
MEP factory credit	–\$2.80/SF	–\$5.40/SF	–\$7.20/SF	–\$9.80/SF
Installed cost/SF wall	\$33.40/SF	\$39.20/SF	\$48.60/SF	\$52.60/SF

Element	T1 — Code+	T2 — Zero	T3 — Positive	T4 — Harvest
Savings vs stick-frame (with MEP)	\$5.10/SF (13%)	\$13.60/SF (26%)	\$22.80/SF (32%)	\$32.00/SF (38%)

Cost basis note: Stick-frame equivalent costs represent the true installed cost of a code-equivalent assembly with equivalent performance. MEP factory credit reflects SIPA data (35–50% field labor reduction on modular MEP assemblies). Volume pricing requires executed supply agreements. All figures ROM ±25%. See [M4Di_ForwardSystem_AssemblyCost.html](#) interactive tool for live scenario modeling.

9. Factory Assembly Enhancement Program

The following table defines recommended factory-assembled components beyond the structural wall panel system — covering fenestration, hardware, fixtures, finishes, and mechanical systems. Each item is evaluated for hub pre-assembly feasibility, tier applicability, and estimated labor and material savings. The Wilmington Hub warehouse program enables volume purchasing leverage across all categories.

Organization: items are grouped by building system. 'Hub pre-assembly' means the work is done at the Wilmington Hub facility before panel or module ships to site. 'Site connect' means the site crew performs only the final connection — no layout, no fabrication, no trade return visits.

Category	Component / Product	T1	T2	T3 / T4	Factory assembly benefit + estimated savings
FENESTRATION — Windows, Doors, Glazed Openings					
Windows	Crystal Magnus 4500/4700 — pre-glazed, pre-sealed unit	Hub-set into pre-framed SIP rough opening · site connect sill tape only	Triple-pane unit pre-installed in panel at hub — arrives sealed	4700 PH unit — factory-set + SIGA FENTRIM IS pre-taped at perimeter	Eliminates window installation trade on site. 2–3 hrs/window saved. Airtight guarantee vs. site-glazed.
Ext. doors	Magnus 4500/4700 swing + tilt-slide — pre-hung	Pre-hung in door buck · factory set in panel · site connect threshold only	Same + triple-pane unit pre-set	PH door pre-hung + sill pan pre-formed in panel	Sill pan failure is the #1 door air leak source. Factory-controlled = warranty-defensible.
Int. doors	Pre-hung wood or HDF — standard size matrix	Pre-hung 3'x8' solid-core, Schlage B-series hardware pre-installed	Same	Same + acoustic door T3 demising walls	Bundle 8–10 door units per order. Hardware pre-installed. 45 min/door site labor eliminated.
Ext. glazing	Fixed lites + transoms — pre-glazed unit	Fixed lites pre-glazed + taped into SIP panel at hub	Same + triple-pane	Same	Eliminates glazier trade. Pre-glazed units airtight-tested at hub before shipping.
HARDWARE — Rough + Finish					
Structural connectors	Simpson MASOZ, HTT hold-downs, SCLC stair cassette	Pre-installed at hub on panel — site: lower panel onto pre-set anchors	Same	Same + A4 SS coastal spec T3+	MASOZ eliminates post-panel field drilling. HTT pre-clipped = 30 min/shear wall saved.
Rough hardware	Joist hangers, blocking brackets, hurricane ties	Pre-clipped at hub with SDS screws	Same	Rtb ALUMEGA pre-installed on CLT element at hub	All rough hardware factory-set. Prevents site-substitution errors.
Finish hardware	Door levers, hinges, strike plates — Schlage B-series	Bundle pre-installed on doors at hub. Same SKU all units.	Schlage Encode Plus smart lock — pre-configured per unit	Same + keyed-alike building master system	Volume purchase 20+ units: ~15–18% below retail. Factory install = no site hardware punch list.

Category	Component / Product	T1	T2	T3 / T4	Factory assembly benefit + estimated savings
Cabinetry hardware	Pulls + hinges — Blum concealed hinges standard	Pre-installed on cabinet doors at hub	Same	Same — Blum soft-close all	Blum Clip Top concealed hinge — factory-set. Eliminates door-adjustment punch list item entirely.
FIXTURES — Plumbing, Electrical, Lighting					
Plumbing fixtures	Kohler or equal — toilet, lavatory, kitchen sink, tub/shower	Prefab wet wall module: PEX manifold + supply stubs + drain rough-in factory assembled and pressure-tested. Site: connect toilet, lav, sink to pre-run stubs.	Same + HPWH pre-positioned in mechanical chase	Same + cistern preheat connection stub T3+	Prefab wet wall eliminates 60–70% of plumber field time. Pressure test at hub = zero leak callbacks.
Kitchen sink + faucet	Kohler Vault or equal — undermount + pull-down faucet	Pre-mounted in countertop at hub — site: drop in + connect PEX stubs	Same	Same	Undermount pre-mounted = no site template work. Supply stubs align to pre-run manifold.
Tub/shower	Kohler or equal fiberglass unit — 1-piece	1-piece unit pre-set in bathroom module before wall close — must be first-in sequence	Same	Tile-ready pan T3+ — pre-waterproofed at hub	1-piece unit requires pre-placement. Sets bathroom module sequence. Hub controls sequence.
Electrical fixtures	Pre-wired J-box locations standard matrix	SIP panel chases pre-routed at hub per standard electrical plan. J-box locations CNC-cut in OSB. Site: pull wire + connect. Eliminates layout step.	Same + SPAN Main 40 pre-wired sub-panel stub	Same + island-mode load schedule pre-configured	Pre-routed chases eliminate electrician layout time (~4 hrs/unit). Reduce NM cable waste.
Lighting	LED can lights — 4" recessed, standard matrix	Pre-wired J-boxes at standard ceiling grid. Fixture drops site-installed.	Lutron Caseta smart dimmer pre-wired	Same + occupancy sensors T3+	Pre-wire at hub + Caseta = compatible with SPAN circuit control. No electrician return visit.
Bath exhaust	Broan or Panasonic — exhaust-only T1 / ERV-connected T2+	Pre-ducted through SIP at hub — site: connect fan unit only	Pre-connected to Vitoair ERV duct run	Same	Pre-ducted exhaust chase eliminates most on-site HVAC rough-in labor for baths.
FINISHES — Interior + Exterior					
GWB / drywall	½" Type X — factory hung + screwed at hub	Hung + screwed at hub. Site: tape + mud + paint only. Eliminates layout + hanging labor.	Same	Same	Estimated 8–12 hours/unit drywall hanging labor eliminated. Screw pattern quality-controlled at hub.
Interior paint	Benjamin Moore Chantilly Lace OC-17 — standard T1	Site-applied 2-coat after mud. Standard color reduces touch-up coordination.	BM HC-170 Revere Pewter accent walls per plan — pre-keyed	Same or BM Regal Select eggshell	Single-color standard = bulk paint purchase. 20-unit order: ~18–22% below retail.

Category	Component / Product	T1	T2	T3 / T4	Factory assembly benefit + estimated savings
Flooring	LVT — COREtec Plus Enhanced or equal · 6"×48"	Pre-cut to room dimensions at hub. Site: float + click-lock. Subfloor pre-leveled.	Same + Schluter DITRA uncoupling under bath tile	Tile bath floors T3+ — Schluter DITRA + 12×24 porcelain · prefab mat cut at hub	Pre-cut LVT eliminates layout waste (~12% standard cut waste → <4% factory-cut). Schluter mat pre-cut at hub.
Kitchen cabinets	Semi-custom RTA — Fabuwood Allure Fusion or equal	Assembled at hub. Pre-installed in cabinet module. Site: set module + countertop.	Same + soft-close Blum all drawers/doors	Same + under-cabinet LED pre-wired	RTA assembly at hub eliminates site assembly time (~6 hrs/kitchen). Module set = 45 min on site.
Countertops	Quartz — Silestone Eternal White or equal · 3cm	Fabricated to kitchen module template. Site: set + silicone.	Same	Same + waterfall end T3+ optional	Template from hub-assembled cabinet module = exact fit. Eliminates field template visit.
Tile — bath	12×24 porcelain — standard T1 bath	Pre-cut tile sets per bath (waste <5%). Site: set + grout. Layout confirmed at hub.	Same + prefab niche box at hub	Dekton or Calacatta porcelain T3+	Pre-cut tile per bath room = zero layout waste. Pre-formed niche = watertight.
Ext. cladding finish	Allura Spectrum pre-finished — factory applied	Factory color no field painting. 50-yr warranty maintained.	Same — full Spectrum palette	Same	Eliminates exterior painting trade entirely. Warranty intact. ~\$1,200–\$1,800/unit paint labor saved.
MECHANICAL + ENERGY SYSTEMS — Factory-Ready Assemblies					
HVAC rough-in	Refrigerant lineset + line-hide	Pre-run lineset in SIP chase at hub. Site: connect indoor + outdoor units only.	Quilt 3-zone: pre-run 3 linesets in chase	Mitsubishi H2i: pre-run + pressure-test at hub	Pre-run + pressure-tested lineset = no HVAC leak callbacks. 3–4 hrs HVAC rough-in saved/unit.
ERV ducting	Vitoair FS-ERV balanced duct system	Pre-ducted balanced supply + return in SIP at hub T2+	Same — PHIUS duct layout	Same	Pre-ducted ERV system = no field duct layout. Eliminates duct leakage test failures.
PV conduit + combiner	EMT conduit + junction boxes for PV	Pre-run EMT + J-box for PV string at hub T2+. Site: pull wire + mount panels.	Same + Enphase trunk cable pre-run	Same + agrivoltaic trellis conduit T4	Pre-run conduit eliminates electrician PV rough-in visit. Saves 4–6 hrs/unit electrician time.
Smart panel stub	SPAN Main 40 pre-wire	200A stub pre-wired at hub T1	SPAN Main 40 load schedule pre-configured to unit plan	Island-mode priority schedule pre-loaded · EcoFlow BMS stub T4	Factory SPAN configuration = no field programming. Island-mode load schedule tested at hub.
Water heater position	HPWH — 50-gal standard	Mechanical chase pre-framed + pre-drilled for HPWH condensate drain at hub	Same + solar thermal preheat stub	Same + cistern preheat loop stub T3+	Pre-framed chase eliminates HPWH coordination on site. Condensate pre-plumbed.
VOLUME PURCHASING — Wilmington Hub Warehouse Program					
SIP panels	ACME 6.5" Neopor GPS	Retail ~\$9.80/SF	Hub volume target: 10% discount	Hub volume target: 12% discount	At 20 units/yr: ~\$14,400 annual savings on SIP alone. Volume = leverage.

Category	Component / Product	T1	T2	T3 / T4	Factory assembly benefit + estimated savings
Allura cladding	Fiber cement lap/panel/shakes	Retail ~\$7.50–\$13.20/SF	Hub bulk: 22–28% below retail	Same + Spectrum pre-finished bulk run	At 20 units: ~\$18,000–\$24,000 annual savings. Largest single bulk opportunity.
Crystal windows	Magnus 4500/4700	Retail ~\$420–\$720/unit	Multi-unit developer pricing: 14–16%	PH order: 16% or better	20-unit window order: ~\$16,000–\$23,000 savings. Crystal national sales rep relationship required.
Connectors (SST+Rtb)	MASOZ, HTT, SCLC, ALUMEGA, HBS	Retail ~\$1.40–\$2.80/SF wall	Contractor direct account: 12–15%	Same	Annual savings modest per SF but consistent. SST contractor account = catalog pricing access.
HVAC systems	Quilt 3-zone, Mitsubishi H2i, Vitoair ERV	Retail standard	Volume contractor: 10–14%	Same	Mitsubishi Diamond Contractor status required (Tier 3). Quilt volume pricing available direct.
Appliances	Refrigerator, range, dishwasher — Builder-grade Samsung or equal	Retail	Builder account: 18–24%	Same + upgrade Samsung Bespoke panel-ready T3+	Builder account + same-spec all units = 20%+ below retail. ~\$2,400–\$3,600/unit savings.

10. Recommended Factory Efficiency Enhancements — Priority List

The following 16 enhancements are ranked by implementation priority and estimated impact. Each is designed to be adopted incrementally as the Wilmington Hub ramps from Phase 1 (assembly + staging) to Phase 2 (full factory-line production). All can be implemented with existing ICC-ES listings, no new testing required.

1. DOOR + WINDOW PRE-INSTALLATION (all tiers)

Pre-install Crystal Magnus windows in SIP rough opening at Wilmington Hub before panel ships to site. SIGA FENTRIM IS airtight tape pre-applied at perimeter. T2+: triple-pane units arrive site-ready. Eliminates window installation trade and guarantees airtight performance. Estimated labor savings: 2–3 hrs/window × 8 windows = 16–24 hrs/unit. Airtight liability shifts to hub QC, not site crew.

2. PRE-HUNG DOOR BUNDLE (all tiers)

Pre-hang all interior doors (3'×8' solid-core) with Schlage B60N hardware at hub. Bundle 8 doors per unit. Exterior doors pre-hung with Magnus unit + sill pan pre-formed. Site crew sets door unit in pre-framed rough opening — no hardware installation on site. Estimated: 45 min/door × 10 doors = 7.5 hrs/unit eliminated.

3. PREFAB WET WALL MODULE — PLUMBING (all tiers)

Assemble PEX manifold + supply stubs + DWV rough-in on a pre-framed 2×4 module at hub. Pressure-test at hub before shipping. Site crew bolts module in place, connects toilet/lav/sink to pre-run stubs. Eliminates 60–70% of field plumber time. Reduces callback risk. Estimated: 12–16 hrs field plumber time → 3–4 hrs site connection only.

4. DRYWALL HANG + SCREW AT HUB (all tiers)

Hang and screw ½" Type X GWB on all interior wall panels at hub. Site: tape + mud + paint only. Eliminates layout + hanging trade from site entirely. Panels arrive site with GWB attached. Estimated: 8–12 hrs/unit drywall hanging eliminated. Quality-controlled screw pattern = consistent fire-rated assembly.

5. PRE-ROUTED SIP ELECTRICAL CHASES (all tiers)

CNC-cut electrical chase locations in SIP OSB face at hub per standard electrical plan matrix. Pre-route conduit or pull string. Site electrician pulls wire + installs devices — no layout. Estimated: 4 hrs/unit electrician layout time eliminated. NM cable waste reduced by pre-measured runs.

6. CABINET + COUNTERTOP MODULE (all tiers)

Assemble RTA kitchen cabinets at hub with Blum hinges + soft-close installed. Deliver as pre-assembled module. Template countertop from module — fabricate quartz offsite. Site: set module (45 min) + set countertop + silicone. Estimated: 6 hrs assembly + 1.5 hrs template visit eliminated per unit.

7. FLOORING PRE-CUT KIT (all tiers)

Cut LVT planks to room dimensions at hub per finalized floor plan. Box per room, labeled. Site: float + click-lock only. Cut waste <4% vs. 12% site-cut. Estimated material savings: ~\$180–\$240/unit on LVT waste reduction alone. Schluter DITRA mat pre-cut for bath modules.

8. HVAC LINESET PRE-RUN (all tiers)

Pre-run refrigerant lineset through pre-formed SIP chase at hub. Cap + pressure-test at hub. Site HVAC crew connects indoor + outdoor units only. Estimated: 3–4 hrs field HVAC rough-in eliminated. Pre-pressure-test = no leak callbacks.

9. SMART PANEL PRE-CONFIGURATION — SPAN MAIN 40 (T2–T4)

Pre-configure SPAN Main 40 circuit assignments, load priorities, and island-mode schedule per unit plan at hub. Download firmware. Ship pre-configured. Site electrician lands wires — no programming. T4: island-mode priority schedule (HVAC → refrigerator → circuits) tested at hub before ship.

10. ERV DUCT MODULE PRE-RUN (T2–T4)

Pre-run balanced supply + return ERV duct runs through SIP at hub per PHIUS-compliant layout. Cap ends. Site: connect to Vitoair unit only. Eliminates duct layout and reduces duct leakage test failures. PHIUS-grade duct layout QC-controlled at hub.

11. ROOFTOP MODULE PRE-ASSEMBLY — LIVEROOF + CROP CIRCLE (T2–T4)

Pre-assemble LiveRoof Standard tray modules at hub for T2 — ship stacked, labeled by roof zone. T3 MAXX: pre-cut RoofBlue risers + MAXX modules. T4: Crop Circle modular planters pre-assembled + ballasted. Site: place per layout plan. Estimated: 4–6 hrs rooftop labor saved per unit.

12. ALLURA SPECTRUM BULK COLOR RUN (all tiers)

Select single Spectrum pre-finished color per development phase. Order all Allura cladding, trim, soffit in one factory color run. Eliminates field painting trade entirely. 50-yr warranty preserved. Estimated savings: \$1,200–\$1,800/unit painting labor + \$200–\$400/unit primer/paint materials.

13. APPLIANCE BUNDLE — BUILDER ACCOUNT (all tiers)

Establish Samsung or LG builder account for same-spec refrigerator, range, dishwasher across all units. Single delivery per cluster. Hub staging. Builder account pricing: 18–24% below retail. Estimated savings: \$800–\$1,200/unit. T3+: panel-ready Bespoke upgrade at similar net cost to standard appliances.

14. KEYED-ALIKE MASTER SYSTEM (all tiers)

Schlage B-series keyed-alike per development phase. One master key. Site manager controls access during construction. Homebuyer receives unit key + master release at closing. Factory-keyed at hub = no locksmith visit. Cost: ~\$40–\$60/unit premium vs. individual keying, saves \$180+ locksmith call.

15. PV CONDUIT + TRUNK CABLE PRE-RUN (T2–T4)

Pre-run EMT conduit + Enphase IQ8 trunk cable in attic/SIP at hub for PV string layout. Site: mount modules + pull string cable. Estimated: 4–6 hrs electrician PV rough-in eliminated. Aligns with SPAN pre-configuration for immediate circuit commissioning.

16. WINDOW + DOOR HARDWARE MASTER BUNDLE

Specify Schlage Encode Plus smart lock for all exterior doors T2+. Factory-program each lock to unit address at hub. Homebuyer activates via app at closing. Eliminates lock swap at move-in. Bulk order 20+ units: ~15–18% below retail. Building manager app = master access during construction.

11. Factory Assembly Savings Summary — Per Unit

The following table aggregates estimated labor and material savings from the full enhancement program. Labor savings are expressed in field trade hours eliminated per unit. Material savings reflect waste reduction and volume purchasing benefits. Figures are ROM ±25% based on SIPA prefab labor data, industry benchmark waste rates, and published volume discount ranges.

Enhancement category	Est. labor saved/unit	Mat. savings/unit	Applies to tiers	Primary dependency
Window + door pre-installation	16–24 hrs	—	All	Hub tooling + SIGA tape stock
Pre-hung door bundle	7–8 hrs	—	All	Schlage bulk account
Prefab wet wall module	12–16 hrs	\$400–600 (callbacks)	All	Hub framing + pressure-test rig

Enhancement category	Est. labor saved/unit	Mat. savings/unit	Applies to tiers	Primary dependency
Drywall hang at hub	8–12 hrs	—	All	Hub panel tilter + crew
Pre-routed SIP electrical chases	4–5 hrs	\$80–120 (cable waste)	All	CNC router at hub
Cabinet + countertop module	7–8 hrs	—	All	RTA assembly station at hub
Flooring pre-cut kit	2–3 hrs	\$180–240 (waste)	All	Panel saw at hub
HVAC lineset pre-run	3–4 hrs	—	All	Pressure-test station at hub
SPAN pre-configuration	1–2 hrs	—	T2–T4	SPAN cloud account + firmware
ERV duct module	3–4 hrs	—	T2–T4	SIP chase routing at hub
Rooftop module pre-assembly	4–6 hrs	—	T2–T4	Staging area at hub
Allura Spectrum bulk color run	6–8 hrs (no paint trade)	\$1,200–1,800	All	Spectrum color decision pre-dev
Appliance bundle — builder account	1–2 hrs (delivery)	\$800–1,200	All	Samsung/LG builder account
Keyed-alike master system	2–3 hrs (no locksmith)	\$140–160	All	Schlage factory keying
PV conduit pre-run	4–6 hrs	—	T2–T4	EMT + trunk cable stock at hub
TOTAL ESTIMATED SAVINGS PER UNIT	80–120 hrs labor	\$2,800–\$3,900 mat	All tiers	Wilmington Hub operational Q2 2027

Bottom line: A fully-operational Wilmington Hub implementing all 16 enhancement categories eliminates an estimated 80–120 field trade hours per unit and \$2,800–\$3,900 in material waste and retail-pricing premiums. At 20 units per year, that is 1,600–2,400 labor hours and \$56,000–\$78,000 in material savings annually — before volume purchasing discounts on SIP panels, cladding, windows, and connectors are applied. Those discounts add an estimated \$38,000–\$62,000 annually at 20-unit production scale.

Implementation gate: The hub pre-assembly program does not require the full manufacturing facility. Phase 1 of the Wilmington Hub (target Q2 2027) — a leased 8,000–15,000 SF light-industrial facility with panel assembly tables, a pressure-test rig, a panel saw, and a CNC router — enables all 16 enhancements. Claymont AE24-002 G4 closeout (Q4 2026) is the prerequisite: it validates the assembly sequence, confirms the connector kit, and populates the ALDDB with actual unit costs that feed the hub production model.